

Analyzing Citizen Sentiment on Public Sector Issues in Sri Lanka Using Machine Learning on YouTube Comment Data (2020-2025)

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Abstract

The social media has become an important platform of social discussion as citizens are able to voice their opinions on the governed and the services provided to the people in large proportions. In Sri Lanka, YouTube is the important element in moulding popular opinion especially the commentary of news item and public affairs shows. The paper introduces a machine learning-powered system to process and categorize citizen sentiments written in Sinhalese in You Tube comments on major matters within the Sri Lankan state apparatus, such as the government policy, fuel supply, electricity, taxation, the economic crisis, healthcare, education, and transportation, and the time frame to be used in the analysis is 2020 to 2025. A total of 7,000 comments were gathered in a dataset of 50 Sri Lankan news and public affairs YouTube channels by using the YouTube Data API v3, only Sinhala-script content was left. After text preprocessing and normalisation 2,500 distinct remarks were acquired. Out of them, 1,000 comments were manually annotated onto three categories of sentiment, namely, positive, negative, and neutral. An extraction of features was done using TF-IDF vectorisation with unigrams and bigrams with a maximum of 5,000 features. A Logistic Regression classifier was trained at the 70 percent of the labelled data, and the weight balancing of the classes was applied to overcome severe class imbalance, and accuracy, precision,

recall, F1-score and confusion matrices were used to evaluate the performance of the classifier. The minimum accuracy of the baseline model was 61% with a high prediction bias on the negative majority class. After balancing of the classes the total accuracy rose to 65 percent, neutralizing class memory rose to 0.39 out of 0.11, and positive class memory rose to 0.64 out of 0.09 resulting in a macro F1-score of 0.60. The trained model was then used to predict the sentiment of the rest of the 1,500 unlabelled comments in the entire dataset. On the overall dataset of 2500 comments, it was found that the negative sentiment prevailed with the majority of 67.3, with the next proportion being 27.1 and 5.6 respectively, which was highly indicative of the socioeconomic sufferings that the people of Sri Lanka had gone through due to the 2022 economic crisis. The article shows that automated citizen sentiment tracking based on social media is possible in a low-resource multilingual South Asian setting and provides practical implications on how to enable evidence-based governance in the public sector in Sri Lanka.

Key words

රජය, ඉන්ධන, විදුලිය, බදු, ආර්ථික අර්බුදය, සෞඛ්‍ය සේවාව, අධ්‍යාපනය, ප්‍රවාහන, Sentiment Analysis, YouTube Comments, Sinhala NLP, Machine learning.

1. INTRODUCTION

The social media sites have essentially changed the way citizens interact with the government and other societal institutions. In the modern democratic society, the freedom of common citizens to illustrate their satisfaction or dissatisfaction of the administration, policies, and provision of services publicly has ceased to be restricted to the traditional methods of elections, petitions, or the mainstream media. Rather, online places, specifically video-sharing platforms like YouTube have become the new space of civic communication, with massive communities being built around shared interests, complaints, and demands of responsibility.

The case of Sri Lanka provides a very strong and pressing reason to investigate this kind. The nation was going through one of the worst economic crises it has had since freedom. Mass fuel crisis, prolonged power outages, skyrocketing inflation, severe shortage of medicines and necessities, and rising national debt have created general demonstrations, changing governments, and unprecedented social movements. During this, the Sri Lankan citizens came in large numbers to YouTube comment sections to share their feelings about events that were happening, ranting their anger, blaming, demanding change, and at times unleashing restrained hope at government announcements.

Regardless of this abundant and unparalleled flow of data on the public opinion, institutes of the Sri Lankan public sector have no systemized way to harvest, process, and respond to citizen sentiment in social media. The extremely large number of comments as well as the non-formal and multilingual character of online discourse in Sri Lanka, where Sinhalese and English are both spoken in addition to Sinhalese and English code-mixing freely together, makes manual analysis impossible. Consequently, policymakers are mostly out of touch with the dynamic mood of the citizens, which fails to offer important chances of responsive and data-driven governing.

Machine learning (ML) and natural language processing (NLP) can be used to fill this gap. Sentiment analysis the mechanical recognition and disambiguation of subjective opinion in text has

come of age in the last 10 years, starting as a naive lexicon-based method all the way to advanced deep learning systems that can process contextually sensitive and multi-lingual data. Current developments in transformer-based models of language, such as Multilingual BERT (mBERT), have seen it become possible to create strong sentiment classifiers even with low-resource languages like Sinhala.

This research paper will fill this gap by building and testing a machine learning pipeline to categorize and analyse the feelings of the citizens in YouTube Sinhala comments on eight main domains of the public sector in Sri Lanka: government policy, fuel supply, electricity, taxation, the economic crisis, healthcare services, education, and transportation. The empirical corpus is comprised of sinhala comments taken on Sri Lankan news media in the years 2020 to 2025, which makes it possible to trace the development of the public sentiment in the years when the economic crisis was the most unstable.

Research Questions

1. Can machine learning models accurately classify citizen sentiment expressed in YouTube comments related to public sector issues in Sri Lanka?
2. Which machine learning algorithm performs best for sentiment classification on multilingual (Sinhala–English) YouTube comment data?
3. What sentiment trends can be observed in citizen discussions on public sector issues in Sri Lanka?
4. What challenges and ethical considerations arise when analysing citizen sentiment using YouTube comment data?

The research has several different contributions. First, it brings the sentiment analysis research to the unexplored realm of the multilingual social media data in Sri Lanka with its issues related to Sinhala script, code mixing, informal lexicon and inadequate standardization of Sinhalese NLP tools. Second, it offers a comparative assessment of several ML and deep learning models - both conventional classifiers and transformer-based ones - in this linguistic and thematic field. Third, it

provides substantive empirical data concerning the trends in citizen sentiment in the face of a national crisis of acute national interest and has direct implications on the citizen policy and operation of government, including citizen engagement. Fourth, it puts the research in a strict ethical context, touching upon the issues of privacy, ownership of data, and bias in algorithms, as well as the responsible utilization of social media data to govern.

2. Literature Review

2.1. Sentiment Analysis: Principles and Development.

Sentiment analysis, also known as opinion mining, is a subfield of NLP that deals with the identification and extraction of subjective information of textual data - polarity (positive, negative, or neutral) of opinions in a text fragment [1]. The discipline, which started with Pang, Lee and Vaithyanathan in the early 2000s, has expanded tremendously to include both basic lexicon-based techniques and advanced deep learning systems [2]. Classic methods of machine learning based on features, including Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) and classifiers, like Naive Bayes, Support Vector Machines (SVM), and Logistic Regression, were the foundational building blocks of the field and they remain competitive baselines [3]. More recently, transformer-based models like BERT, RoBERTa, and their multilingual counterparts have achieved a new performance standard on a multitude of tasks in NLP [4].

2.2. Sentiment Analysis on You Tube and Social Media.

Social media create text that is of scale and informality that presents challenges to NLP systems. YouTube is one of the most popular video-sharing websites in the world, and its comment sections make up a huge, vibrant database of the opinions of the people [5]. It has been shown that the sentiment analysis of YouTube comments to determine trends and seasonality can offer an unbiased view of how real-world events

can affect the sentiment of the people with the user sentiment trends being highly correlated with the real events in the context of the relevant keywords.

A major problem with sentiment analysis on YouTube is the number of comments that should be processed; further, due to the informal nature of YouTube comments, use of slang and jargon, machine learning algorithms may have a hard time discerning the sentiment of comments. Deep learning models such as Recurrent Neural Networks (RNN) and Gated Recurrent Units (GRU) IJRASET Techniques have been used to analyze YouTube comments with better performance than classical methods [6].

A study that categorizes YouTube comments has explored various sentiment analysis methods, and it has also been determined that the Bi-LSTM architecture with BoW features provides the best settings of hybrid NLP models in working with YouTube comment data. NLP-based model-based studies to classify YouTube comments on Arabic content and established that Naive Bayes reached an accuracy of 94.62, which is an indication that even the old-fashioned classifiers are competitive in non-English contexts [7].

YouTube political sentiment is also of interest to scholars. Survey examining the opinion of the people on political leaders using YouTube and Twitter commentaries has identified the influence of the medium on the context of the statements made by people, and the results of the survey have shown that there are significant differences in the way people express their opinions on the two sites.

2.3. Sentiment Analysis of Public Sector and Governance.

Sentiment analysis as a tool to aid in evidence-based governance has become an area of increasingly popular scholarship interest. Social media has been known to be useful in supporting co creation between citizens and local governments with social listening being ranked better than other conventional approaches such as surveys to capture the voice of the citizen in its natural and authentic form.

A literature review systematically of sentiment analysis implemented in government development policies found SVM, BERT, and Naive Bayes as the most applied and effective algorithms, but

SVM has limited capabilities with structured information and easier tasks, and BERT with better capabilities with nuanced textual information.

Sentiment analysis and topic modelling in the public service are time-tested methods applied to process the comments of citizens in social media, actively promoted by machine learning and deep learning models, but with certain limitations such as contextual relevance, poor performance in low-resource languages, and lack of interpretability.

The studies on the use of social media to govern in the context of developing countries have also revealed that sentiment analysis can be applied in real-time policy responses. A Saudi Arabian educational reforms study created a sentiment analysis application based on AraBERT, which analyses the emotional reactions of the masses on Twitter, and the fine-tuned model has an F1 score of 0.89, showing that the sentiment analysis of the mass on the Twitter feed can offer direct feedback to government institutions in their current and future planning.

2.4. Sinhalese NLP and Low-Resource Linguistics Problems.

The majority language in Sri Lanka is Sinhalese, which poses special problems to NLP research. Sinhalese is resource deprived in the eyes of Natural Language Processing systems and study as it lacks the economic motivation of English and even the sheer volume of languages like Chinese and thus NLP development is scattered and lacking in coordination of research teams.

To perform sentiment analysis in Sinhalese, a study involving deep learning (RNN, LSTM, Bi-LSTM, capsule network) on a Sinhalese news comment corpus showed that although sequence-based models are feasible in a comprehensive manner, the lack of resources and morphological diversities of Sinhalese remains to be a key challenge.

Studies of code-mixed Sinhalese texts (also known as Singlish) have established that standard NLP algorithms have their performance significantly impaired when used on such data. Aspect-based sentiment analysis A hybrid model using XLM-RoBERTa with domain-specific lexicon correction provided 92.3 accuracy and 0.89 F1-score in

English, and 88.4 accuracy and 0.92 F1-score in Sinhala and code-mixed text showed that the transformer-based models are an important improvement over traditional machine learning-based methods in low-resource and code-mixed language processing.

Sinhalese Offensive Language Dataset (SOLD) [8] and Sinhalese-CMCS dataset [9] are both valuable community resources in the construction of annotated data on Sinhalese social media NLP. On the Sinhala-CMCS dataset, 10,000 social media comments were annotated with five classification tasks, such as sentiment analysis, humour detection, hate speech detection, language identification and aspect identification, and it serves as a baseline resource to multi-task Sinhala NLP studies.

2.5. Feature Extraction: TF-IDF and N-Gram Methods.

TF-IDF is also one of the most popular features extraction methods used in text recognition. A comparative analysis of TF-IDF weighting schemes revealed that TF-IDF has the highest accuracy, precision, recall and F1-score value when used with n-gram features in isolation and trained on unstructured sentiment data using classifiers of SVM, Logistic Regression, Multinomial Naive Bayes and Random Forest.

Bigrams together with the unigrams in the TF-IDF vectorisation have been demonstrated to elicit significant contextual co-occurrences unattainable by unigrams, especially in agglutinative morphology languages such as Sinhalese [10]. Logistic Regression TFIDF has been determined to be a powerful yet understandable baseline in sentiment classification tasks. Support vector machines and logistic regression are classical approaches to machine learning that are powerful text classification methods, and even though they are frequently replaced by deep learning techniques, have consistently high performance on a wide range of data sets.

2.6. Machine Learning models on Sentiment Classification.

Optimally, Logistic Regression is an established multi-class text categorizer. Regardless of the deep learning resources available to supply state-of-the-art results, it has always provided competitive results when trained on TF-IDF features, especially when the labelled datasets are small [11]. Twitter sentiment analysis TF-IDF Logistic Regression studies have seen that the pipeline provides a convenient and reproducible opinion mining baseline in real-time with the primary drawbacks being inability to work with short texts, sarcasm, domain shift, and class imbalance [12].

SVM based classifiers have shown good performance in a variety of sentiment benchmark datasets. Naive Bayes is a simple classification algorithm though it is also an effective and a fast classifier especially in high dimensional text features space. Random Forest models offer the benefit of being ensemble-based that is robust but need bigger datasets to beat those of linear classifiers in text domains [3].

The common practice of class imbalance, that is inherent in real-world social media sentiment datasets where negative remarks are common, is normally overcome with cost-sensitive learning, class weight modifications, or oversampling algorithms like SMOTE [13]. In the present paper, the built-in parameter of the Logistic Regression of `class_weight = balanced` was used to balance out the extreme unequal distribution of the labelled dataset.

2.7. Deep Learning and Transformer-Based Approaches.

The implementation of BERT (Bidirectional Encoder Representations into Transformers) by Devlin et al. [14] has changed the scope of NLP in a fundamental way. The pre-training on large corpora and the fine-tuning paradigm of BERT has provided state-of-the-art performance on all benchmarks of NLP. Its multilingual version, mBERT, which is pre-trained on 104 languages, has generalized these abilities to low-resource languages.

A study on multi-class sentiment analysis of Urdu text with multilingual BERT has shown that the

proposed mBERT model with BERT pre-trained word embeddings was superior to deep learning, machine learning and rule-based classifier and the F1 score of 81.49 indicates the effectiveness of mBERT in sentiment classification of low-resource language using deep learning methods. Nature

An overview of multilingual sentiment analysis methods identified that transformer architectures are consistently resource efficient on underrepresented languages, and multilingual BERT is very accurate on languages like Kurdish whereas hybrid models consisting of Naive Bayes and Logistic Regression exhibited competitive results on languages like Serbian.

Sequential text data have also been shown to be effectively represented in Long Short-Term Memory (LSTM) networks especially when there are contextual dependencies over longer sequences that matter [15]. In the case of Sinhala in particular, LSTM, Bi-LSTM models have shown the best performances before the emergence of transformer models [16].

2.8. Social Media Data Analysis Ethics.

There are significant ethical issues that are raised by the gathering and processing of social media information. The researchers will be forced to balance the problem of informed consent, privacy of the users, anonymisation, and terms of service of the platform [17]. The public comment data of YouTube is usually publicly accessible to research, having the terms of API usage of the platform, although anonymisation of usernames, safe data storage, and non-individual identification are advised best practices [18]. Another issue is algorithmic bias: when a training sample contains a set of data that biases specifically towards a particular group of people, dialect, or perspective, the model trained on it can make biased or unfair predictions [19]. The problem of representation bias is significant in the framework of the Sri Lankan discourse because the data will be composed solely of Sinhala-script comments, which might not strongly reflect the opinion of Tamil-speaking or English-speaking groups [20].

3. Methodology

3.1 Research Design

The research design was based on a data-driven, quantitative, experimental study where it analysed the sentiment of the citizens in relation to the issues in the public sector in Sri Lanka through machine learning methods. The study will be based on the Cross Industry Standard Process of Data Mining (CRISP-DM) model that gives a systematic method of data analysis and model creation. There are several steps in the workflow: data collection, data preprocessing, data annotation, feature extraction, model development, evaluation and sentiment analysis.

This methodology aims to turn disorganized data in form of YouTube comments into valuable information using systematic process and machine learning. The design also provides reproducibility, scalability, and reliability of the results.

3.2 Data Collection

The YouTube Data API v3 was used to collect data programmatically based on the publicly available comments. The total number of Sri Lankan news channels and public affairs shows that were sampled in the study was 50 YouTube videos published between 2020 and 2025. These were selected videos that were related to major issues in the public sector.

The predetermined keywords that were used to guide the selection process included fuel shortages, electricity supply, healthcare services, education policies, taxation, and transportation as the major national issues. The topics were chosen as they greatly affect the lives of the citizens and were discussed in the press at the time of study.

The original set of data comprised about 7,000 comments. Only the comments with Sinhala characters were kept in order to keep the focus on the local sentiment. This step of filtering was made to make sure that the dataset was closer to the real opinions of citizens of Sri Lanka. All privacy information like usernames and timestamps were omitted to keep the data private and concentrate on the text

3.3 Data Preprocessing

Preprocessing of textual data is among the most important stages of raw data preparation in machine learning. The comments that are made on YouTube are noisy with informal language, emojis and URLs and other non-textual content. Thus, Python and regular expression methods were used to perform a lot of cleaning.

The preprocessing stages involved:

- Deletion of URLs, emoticons, special characters, newlines, and tabs.
- Change of text to lowercase to maintain consistency.
- Whitespace normalization and normalization of punctuation.
- Elimination of repeated remarks to prevent bias.
- Deletion of blank or null records.

Furthermore, the type of comments that were filtered was to include the Sinhalese characters and to allow basic English ones where required. This was significant in the management of code-mixed language patterns.

The dataset was processed and narrowed down to 2,500 clean and useful comments out of 7,000 comments. This purged data formed a sound base to be used in further analysis.

3.4 Data Annotation

To train a supervised machine learning model, it is necessary to have labeled data. Hence, 1,000 comments were annotated using the manual method to three sentiment groups:

- Positive
- Negative
- Neutral

Annotation was done by human beings to provide emotional interpretations of every comment. In order to maintain uniformity, the rules of labeling were specified, with a definition of each sentiment category. As an example, comments that gave out dissatisfaction or criticism were considered as negative, and comments that gave out approval or

satisfaction were considered as positive. Neutral statements comprised factual statements or vague opinion.

The programmatically corrected common labeling errors (e.g. positive, neutral) were rectified to normalize the dataset. In spite of the fact that inter-annotator agreement was not calculated formally, cross-verification was performed in order to enhance the reliability of labeling.

The other 1, 500 comments were not labeled and are used subsequently to predict and analyze trends.

3.5 Feature Extraction

The field of transforming textual information into machine learning-accepted numbers is known as feature extraction. In the present research, Term Frequency-Inverse Document Frequency (TF-IDF) vectorization was applied.

TF-IDF contains the significance of words in a document compared to the whole set of documents. The words that are common in a particular comment and are not so common in all comments are assigned higher weights, and consequently, they are more informative in classification.

The configuration that was applied was the following:

- Word combinations and use of unigrams and bigrams to include individual word and word combinations.
- Dimensionality control by maximum feature limit of 5,000.

The given approach led to the sparse matrix representation of the data, in which every comment was converted into a numerical feature vector. TF-IDF is also strong in text classification because it can identify discriminating words.

3.6 Model Development

The main machine learning model that has been chosen in this study is the Logistic Regression classifier. The use of the Logistic Regression is popular in text classification because it is easy and has an interpretive effect and can be used when the data is high-dimensional.

Three subsets were created out of the labeled dataset:

- 70% for training
- 15% for validation
- 15% for testing

The dataset had a major problem of class imbalance because the negative comments outnumbered the positive ones. To deal with this, weights on classes were used in training the models. This was to make sure that minority classes (positive and neutral) were given higher significance, which enhanced the capacity of the model to learn using imbalanced data.

A feature matrix in the TF-IDF format was used to train the model and hyperparameters were optimized on the validation set. Joblib was next used to save the trained model to use later.

3.7 Model Evaluation

The multiple metrics were used to evaluate the performance of the model and provide the comprehensive assessment:

- Accuracy: Assessment of general accuracy of predictions.
- Precision: Refers to the percentage of all accurate positive predictions.
- Recall: Tests the ability to recall all the relevant instances.
- F1-score: Precision and recall arithmetic mean.
- Confusion Matrix: Gives a breakdown of results in classification.

The model was also cross validated to check the robustness and the generalizability of the model. It is a method to mitigate the overfitting risk and to have a more accurate idea of the model performance.

The trained model was subsequently applied to the 1,500 unlabelled comments to generate predicted sentiment labels, enabling a comprehensive analysis of sentiment distribution across the complete dataset

The results were predicted and then summed up to locate the general sentiment distribution. Also,

topic-based analysis was performed to analyze the patterns of sentiment in various issues in the public sector. Findings were presented through visualization techniques which included bar charts and pie charts.

3.8 Ethical Considerations

The concept of ethics was wisely considered during the research. All publicly available YouTube comments were used which guaranteed that the policies of the platform were observed. The user identities were not identified, and no personally identifiable information was held.

4. Results

4.1 Summary of Experimental Results

This part shows the results of the sentiment classification model that was made for this study. The analysis concentrates on the labelled dataset employed for training and evaluation, in addition to the unlabelled dataset utilised for extensive sentiment prediction. We use several different metrics to judge how well the machine learning model works, and we get insights from the way the sentiment classes are spread out.

The results show how well the model works, how the classes are spread out, and how people's feelings about public sector issues in Sri Lanka are changing over time.

4.2. Baseline Model Performance

The first Logistic Regression model was trained on 70 percent of 1,000 labelled comments (with no balancing of the classes). The classification report on this baseline model on the 280-comment test set is shown in Table I.

Accuracy: 0.6107142857142858				
Classification Report:				
	precision	recall	f1-score	support
negative	0.64	0.91	0.75	176
neutral	0.31	0.11	0.16	82
positive	1.00	0.09	0.17	22
accuracy			0.61	280
macro avg	0.65	0.37	0.36	280
weighted avg	0.57	0.61	0.53	280

Table.1.

The initial model had an overall accuracy of 61.07. Although the negative class had quite a high recall of 0.91, the neutral and positive classes had an extremely low recall of 0.11 and 0.09 respectively. Though the positive class had a precision of 1.00, this is deceiving because it simply indicates that on the infrequent instances that the model predicted positive, it was correct, but it hardly ever predicted positive at all as indicated by the close-to-zero recall of 0.09. The macro-average F1-score of 0.36 only proves that the discriminatory ability of the model in all three classes as a block was poor.

This behaviour is a direct result of the drastic class imbalance of the labelled dataset in which the negative comments constituted 89.1 percent of all annotations. The model picked up an almost constant prediction strategy to go to negative on most inputs, and this was a high-accuracy predictor of the majority input class but had no ability to predict the sentiment classes that formed the minority. The finding highlights the importance of balancing classes prior to the implementation of any sentiment classifier on imbalanced social media data of real-life.

4.3 Improved Model Performance (Class balancing)

To achieve the imbalance that was observed during the baseline evaluation, the Logistic Regression classifier was retrained using the class weight=balanced parameter. In this setup the model is told to pay heavier penalties on cases of misclassifications of minority classes in the training process which is worked out as re-weighting the loss function proportional to the inverse class frequencies. The classification report of this improved model on the same 280 comment test set is shown in Table II.

Classification Report:				
	precision	recall	f1-score	support
negative	0.74	0.77	0.76	176
neutral	0.44	0.39	0.41	82
positive	0.61	0.64	0.62	22
accuracy			0.65	280
macro avg	0.60	0.60	0.60	280
weighted avg	0.64	0.65	0.64	280

Table. II

The class-balanced model showed significant and significant gains in all the minority sentiment classes. The total accuracy was improved to 65 percent. More importantly, the neutral class recall changed to be stronger (0.11) to 0.39, and the positive class recall changed to be stronger (0.09) to 0.64, which is more than seven times the change. There was an increase in the macro-average F1-score by a factor of 0.36 to 0.60, which means that the model is currently performing reasonably and consistently with respect to the three sentiment classes as opposed to being compelled by a single class.

It is also important to note that the negative class recall was reduced marginally in the range of 0.91 to 0.77 due to the balancing. Such a trade-off was anticipated and is acceptable: the model dropped a measure of negative-class recall getting considerably better sensitivity to the neutral and positive classes, which leads to a much more balanced and useful-in-practice classifier.

Figure 1 shows the confusion matrix of the balanced Logistic Regression model being tested on the entire dataset, and Figure 2 shows the confusion matrix of the balance Logistic Regression model being tested on the held-out test set.

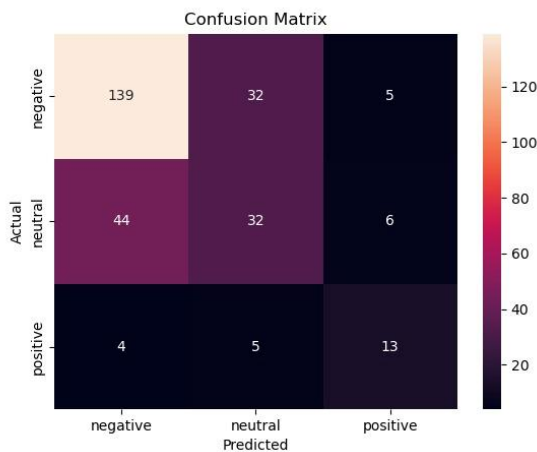


Fig. 1. Confusion Matrix — Balanced Logistic Regression Model

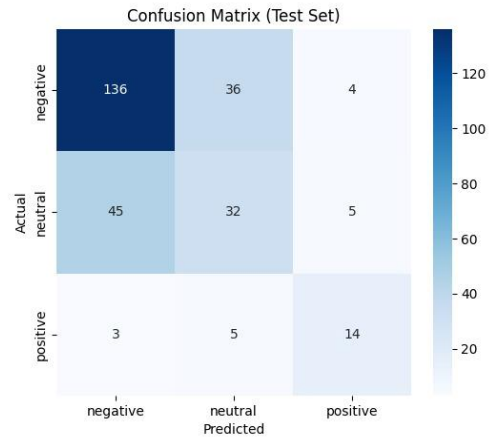


Fig. 2. Confusion Matrix — Test Set (280 Comments)

Based on Figure 1, the model correctly categorized 139 of 176 negative remarks, 32 of 82 neutral remarks and 13 of 22 positive remarks in the generalized assessment. On the held-out test set, in particular, the model managed to identify 136 negative, 32 neutral and 14 positives by the correct model.

The most salient misclassification in the two matrices is the negative prediction of neutral remarks and this takes place 44-45 times in judgments. This tendency can also be explained linguistically: in Sinhala official discourse, even purely informational statements about such events as electricity rationing schedules or fuel queue announcements often contain words that have negative connotation, e.g. “කප්පාදුව” “(cuts/shortages) or “ගැටලුව” (problem), and the distinction between a statement of neutral character and a statement of negative character is unavoidably vague to the token-level classifier. The second most frequent error was negative-to-neutral misclassification (32-36 instances), and the positive comments were represented in the training data very rarely, and thus, 13-14 out of 22 were correctly identified.

4.5. Comprehensive Sentiment Distribution.

After reviewing the classification model, the trained Logistic Regression classifier and trained TF-IDF vectorise were used on the rest of the 1,500 unlabelled comments to produce predicted

sentiment values. The obtained predictions were then added to the 1,000 manually labelled comments to come up with the general sentiment distributions of the entire dataset of 2,500 comments. This distribution is shown in figures 3 in bar chart.

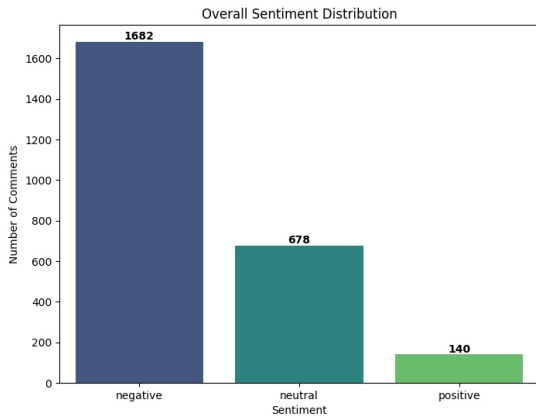


Fig. 3. Overall Sentiment Distribution — Bar Chart (Full 2,500-Comment Dataset)

The most common category in the entire data was negative sentiment (1,682 comments 67.3). The comments made under neutral sentiment constituted 678 (27.1) and positive sentiment made the least at 140 (5.6). It is a distribution of the overall socioeconomic climate in Sri Lanka in 2020-2025 and is most acutely defined by the 2022 economic crisis when the citizens were facing acute shortages of fuel and electricity, in addition to having needed essential medicines. The comparatively large neutral percentage (27.1) relative to the labelled one (8.1) can probably be explained by both the increased sensitivity of the model to the neutral expressions after class balancing and the natural difference between the annotated and non-annotated sections of the dataset. The relatively small percentage of positive sentiment (around 5.6% throughout the entire dataset) supports the fact that approval of the government work on the most popular topics in the state was extremely uncommon during this time, which also correlates with the previously reported magnitude of dissatisfaction among the population that eventually led to the 2022 popular rebellion in Sri Lanka.

One important thing to note about the results is that the full dataset has a much higher percentage of neutral sentiment (27.1%) than the manually

labelled subset (8.1%). This means that the first step in the annotation process may have favoured more extreme expressions of sentiment, especially negative comments. On the other hand, adding machine-predicted labels to the unlabelled data seems to capture a wider range of moderate or informational statements. This shows how important it is to use large-scale automated sentiment prediction to get a more accurate and representative picture of public opinion.

4.6 Key Findings

The findings in this study point to some significant findings. To begin with, the distribution of sentiment is strongly negative, suggesting that there is a high level of dissatisfaction of the people with the issues in the public sector in Sri Lanka over the period of time studied. Second, the poor performance of the baseline model in the minority class detection by the severe class imbalance proved the significance of data balancing techniques. Third, the application of weighted classes did indeed greatly enhance the performance of the model, especially in the neutral and positive sentiments, raising the macro F1-score to 0.60.

Moreover, the analysis of the confusion matrix demonstrated the common misclassification of the comments as negative because of the similarity of linguistic patterns in Sinhalese. Lastly, massive prediction on the entire dataset validated the use of negative sentiment as dominant and replicated the socio-economic realities on the ground. These results show how well, as well as how poorly, machine learning models perform in multilingual sentiment analysis tasks.

The findings suggest that machine learning models, namely, Logistic Regression, can be used to categorize the sentiment of citizens with a moderate precision (65) on Sinhala comment data on You Tube. Although the model is effective in dominant sentiment classes, the minority classes are poor because of imbalance in data and linguistic complexity. Hence, machine learning is a viable technology that needs preprocessing and balancing methods to achieve high efficiency in multilingual applications.

5. Discussion

5.1 Interpretation of Model Performance

The findings of this paper indicate that a Logistic Regression classifier with TF-IDF feature extraction is able to classify citizen sentiment on Sinhala YouTube comment data with moderate but meaningful accuracy. The final balanced model achieved an overall accuracy of 65%, confirming that machine learning approaches can be applied to sentiment extraction in low-resource language conditions, even when labelled data are limited and class imbalance is severe.

The model performed strongest on the negative sentiment class, achieving an F1-score of 0.76. This is directly attributable to the overwhelming presence of negative examples in the training data, which enabled the model to acquire robust discriminative patterns associated with dissatisfaction and criticism toward public sector services. The neutral and positive classes performed considerably weaker in the baseline model due to their significantly smaller representation, with recalls of only 0.11 and 0.09 respectively.

The application of class weight balancing proved to be a critical intervention. Neutral class recall increased from 0.11 to 0.39, and positive class recall improved from 0.09 to 0.64, while the macro F1-score rose from 0.36 to 0.60. Although this came at a minor cost to negative class recall, which decreased from 0.91 to 0.77, this trade-off is entirely justified as it produced a substantially more balanced and practically useful classifier capable of detecting sentiment across all three classes rather than defaulting almost entirely to negative predictions.

5.2 Problems with Multilingual Sentiment Analysis.

The complexity of the analyzing sentiment in Sinhala text is one of the biggest challenges noticed during this research, especially when it is coupled with the code-mixed language trends. Informal expressions, abbreviations, and slang were also common in numerous YouTube comments with the combination of Sinhalese and English words and casual expressions. Such properties of language

render it challenging to use traditional machine learning models to absorb the contextual meaning. The other serious issue is the existence of ambiguous and context-specific expressions. In Sinhalese speech, some words usually connoted to negative cases like words relating to deficits or issues, can be used neutrally and negatively. Subsequently, the model tended to classify neutral comments as negative as the confusion matrix analysis shows.

There were also significant challenges of sarcasm and irony. Bag-of-words or TF-IDF representations give machine learning models that fail to consider deeper contextual information, which makes them make incorrect predictions when comments have implicit or subtle sentiment.

5.3 Comparison with Existing Literature

The results of this paper are in line with what has been found in sentiment analysis research. Past research has indicated that conventional machine learning algorithms like Logistic Regression, Support Vector Machines and Naive Bayes can work reasonably well when used in conjunction with feature extraction methods like TF-IDF.

Nevertheless, like other research, the results of these models are strongly dependent on the nature of the datasets especially the class imbalance and complexities of the language. The accuracy levels usually are higher in research carried on English datasets because of the presence of an organization of annotated corpora and other advanced NLP resources. In its turn, this paper brings out the extra issues of low-resource languages like Sinhala.

The enhancement made following the application of the class balancing techniques is consistent with the results in earlier studies, which indicate that it is essential to take into account the problem of data imbalance in order to obtain fair and credible classification outcomes. In addition, the constraints that have been reported in this paper such as the inability to manage sarcasm and code-mixed text are common issues in multilingual sentiment analysis.

5.4 Understandings of Citizen Sentiment.

In addition to model performance, the findings can be useful in understanding the mood of people on various issues in the public sector in Sri Lanka. The level of dissatisfaction of the people is high because of the preponderance of the negative sentiment in the labeled and unlabeled data.

The trend could be directly associated with the socio-economic situation in Sri Lanka in 2020-25 and, specifically, with the economic crisis of 2022. Problems like fuel shortages, electricity outages, inflation, and lack of access to basic amenities greatly affected the day-to-day life of the citizens, and this has contributed to a lot of frustration and criticism as aired through social media.

The comparatively low percentage of the positive sentiment implies that the popularity of the government initiatives among the population was low. It was common to get positive statements related to certain improvements or temporary relief options than general satisfaction with the services of the people.

Another insight can be given by the fact that the neutral sentiment is higher in the full dataset than in the labeled subset. It indicates that mass-scale automated prediction has a wider scope of capturing more informational or less emotionally strong comments and provides a more balanced view of the discussion of people.

5.5 Practical Implications to the Public Sector.

The results of this research are significant in regard to the governance of the public sector in Sri Lanka. The capability to process sentiment of citizens automatically through social media information is a useful resource to policy makers and government entities.

To begin with, sentiment analysis would make a good real-time monitoring system where upcoming issues and social concerns will be detected. Through constant review of posts on social sites such as YouTube, the government will be in a position to identify patterns of dissatisfaction and take action beforehand.

Second, sentiment analysis can be used to inform evidence-based decision-making. Knowing about

the social perceptions enables the policymakers to determine how effective the policies and services are, where they need to be improved on, and how to allocate the available resources.

Third, sentiment analysis can be used to improve transparency and citizen engagement through machine learning. Governments can instill trust and enhance communication between government and citizens by recognizing and responding to citizens through addressing their issues of concern.

5.6 Limitations of the Study

Irrespective of what it has done, this study has a number of limitations. The set size of the labeled data is not very large, and this limits the capability of the model to follow complicated tendencies, especially in terms of minority sentiment classes. It is probable that an increase in annotated samples would enhance the performance of the model.

The issue of class imbalance is still beckoning despite the use of weighting methods. Although balancing enhances the performance of minorities classes, it can lead to trade-offs in accuracy.

The second weakness is that it targets the Sinhalese only remarks. As this makes it relevant to the local sentiment, it takes out English and code-mixed comments, which are also common in Sri Lanka. The consideration of multilingual processing would give a more thorough analysis.

Also, the analysis is based on conventional machine learning methods that do not necessarily determine the contextual peculiarities of text. Their sophisticated models, like deep learning or transformer-based models, might be able to perform even worse.

5.7 Future Research Directions

This study can be enhanced in future by examining more complicated methods of sentiment analysis. Deep learning models like Long Short-Term Memory (LSTM) networks, transformer-based models like Multilingual BERT may be used to better address the context, sarcasm, and code-mixed language. It would be important to increase the amount of data in terms of annotated comments to increase model training and evaluation. Also, it would be best to include multilingual data such as

English and code-mixed text to obtain a full picture of the malaise in society. The other possible direction is the combination of the topic modelling techniques in order to generate certain themes related to alternative categories of sentiments. This would enable the analysis of the issues of public concerns and policy-related issues in more detail. Lastly, the emergence of real-time sentiment analysis systems may allow the opinion of the people to be followed 24/7 which can be very useful in terms of dynamic and responsive governance.

6. Conclusion

This paper examined how machine learning methods can be used to examine citizen sentiment on matters related to the public sector in Sri Lanka by using YouTube commentary data between 2020 and 2025. Using the tools of natural language processing and a Logistic Regression classifier, the study revealed that it is possible to derive valuable insights out of large amounts of unstructured multilingual social media data. The model had a moderate accuracy of 65, and a high level of correct classification of negative sentiment, which was the prevailing category in data set. Class balancing also helped the model a lot in identifying the minority classes especially the neutral and positive types, which leads to the significance of combating the issue of data imbalance when performing tasks of determination of sentiments. The results have demonstrated that the negative sentiment was dominant in all the discussions in the public, which showed the general unhappiness towards some of the most important concerns in the public sector like fuel shortage, power outages, and economic crises during the research course. In addition to technical performance, the research offers a well-grounded idea of what the population thinks and can show what the idea of machine learning can be used as the aid to the data-driven governance. Through feedback on citizens, the systematic analysis of citizen feedback will help the institutions of the country to understand their concerns, track the emerging issues and make better informed political decisions. Nevertheless, the research also found several limitations such as the size of the labelled dataset used was low, there was class imbalance, and there were difficulties in

working with Sinhala language and informal social media text. These shortcomings indicate that more studies are required based on larger datasets and sophisticated models. To conclude the study, it forms a baseline on which the sentiment analysis can form in the Sri Lankan context and establishes the significance of applying social media analytics in the decision-making process of the population sector.

7. References

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